16.8.1 Problems

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The logit model parameter estimates and their standard errors are

$$\underset{(\text{se})}{\tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME} = \underset{(0.7505)}{-0.2376} + \underset{(0.2064)}{0.5311} DTIME$$

a. Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.

b. Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [*Hint*: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]

c. Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?

d. Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

(a)

z = -0.2376 + 0.5311* 1 = 0.2935

P(AUTO = 1 / DTIME = 1) = 1/{1 + e^{-0.2935}} =57.3%

(b)

z = -0.0644 + 0.3* 1 = 0.2356

Phi(0.2356) =59.31%

(c)

z_1 = -0.2376 + 0.5311 * 30 = 15.6954

P_1 = 1/{1 + e^{-15.6954}}= 1

z_2 = -0.2376 + 0.5311 * 40 = 21.0064 \implies

P_2 = 1/{1 + e^{-21.0064}}= 1.0000

LPM（線性機率模型）的邊際效應

Delta P = beta_{LPM} * Delta DTIME = 0.0703 * 10 = 0.703

兩者結果存在實質且巨大的差異，這正好凸顯了線性機率模型（LPM）在極端值下的致命缺陷。Logit 模型呈現了合理的『邊際效益遞減』特性

(d)

Logit 模型的計算結果
當 DTIME = -50 時：
z_1 = -0.2376 + 0.5311*(-50) = -26.79
P_1 = 1/{1 + e^{26.79}} =0 當
DTIME = -40 時：
z_2 = -0.2376 + 0.5311* (-40) = -21.48
P_2 =1/{1 + e^{21.48}}=0

(probit)
當 DTIME = -50 時：
z_1 = -0.0644 + 0.3* (-50) = -15.06
P_1 = Phi(-15.06)=0
當 DTIME = -40 時：
z_2 = -0.0644 + 0.3* (-40) = -12.06
P_2 = Phi(-12.06)= 0